Alexander Karachun | **4th-year Applied Mathematics and Information Science Student**

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PROFESSIONAL EXPERIENCE

Student Researcher - HSE University / AIRI **2025–Present**

Research project on VLM-guided image generation, conducted at the HSE Faculty of Computer Science under the supervision of Aibek Alanov, a scientist at AIRI.

- Designed and implemented three VLM-guided generation algorithms and the supporting multi-GPU experimentation infrastructure, building on a broad literature review of diffusion guidance and vision-language methods.
- Ran extensive experiments to identify the most effective approach, analyzed its limitations, and prepared the results for publication.

Teaching Assistant - HSE University, Faculty of Computer Science **2025–2026**

Machine Learning 1 course for undergraduate students.

- Prepared course materials and supported students with core machine learning concepts.
- Graded homework for a cohort of 15 students.
- Administered examinations and assessed student submissions.

Olympiad Organization Assistant - Russian National AI Olympiad **2026**

- Assisted in organizing the Olympiad by creating a competition dataset for its organizers.

EDUCATION

HSE University, Moscow **2023–2027**

BSc in Applied Mathematics and Information Science

Fourth-year student; specialization in Machine Learning and Applications

Relevant Coursework

Deep Learning and NLP: Deep Learning; Machine Learning 1-2; NLP, Yandex School of Data Analysis; Optimization Methods for ML; Research Seminar; Hugging Face Agents Course.

Mathematics: Mathematical Statistics; Probability Theory.

Systems: CUDA; Algorithms and Data Structures.

COURSE PROJECTS

- **Self-Made Stable Diffusion 1.5** (GitHub)
  - Implemented the Stable Diffusion 1.5 architecture and inference pipeline in PyTorch, including the CLIP text encoder, VAE, U-Net, attention layers, DDPM sampler, and classifier-free guidance.
  - Built support for text-to-image and image-to-image generation.
- **NeuroPushkin** (GitHub)
  - Built a dataset of approximately 7,000 synthetic texts through the OpenAI API and fine-tuned ruGPT-3 Medium to generate prose in Alexander Pushkin's style.
  - Explored the design space through ablations over model architectures (RNN, GRU, LSTM, ruGPT-3) and tokenizers, including a custom-trained BPE tokenizer.

TECHNICAL SKILLS

ML/DL: **Python, PyTorch, Datasets, Transformers, Diffusers, Hydra**

Tools: **Comet, TensorBoard, SQLAlchemy**

AI Agents: **MCP, smolagents, LangGraph, LlamaIndex**

Systems: **C++, CUDA, Triton, basics of Linux**

PERSONAL INTERESTS

Deeply passionate about deep learning, methodically building expertise area by area – from classical CV and NLP to LLMs, diffusion-based generation (images and text), VLA, and agent harness development – and eager to go deeper into any of these or take on new ones.

Speed typing: reached level 230 on Monkeytype, with a personal best of 129 words per minute.